HDOC - Day 8 Activity

C Programming Assignment

Name-Arushi Gusain

Sap id-590033489

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Question 1:

Write a C program to find the radius, diameter, and volume of a sphere when its surface area is given.

Input, Output and Formula

Input: Surface area of the sphere **(A)**
Output: Radius **(r)** ,Diameter **(d)** & Volume **(V)**

Formula Used:

$$r = \sqrt{\frac{A}{4\pi}}$$

$$d = 2r$$

V=(4/3)*π*r^3

Standard Test Case Table

Test case	Input(A)	Expected output(r, d, V)	Actual output	Result
1	314.16	5,10 ,523.60	Same	Pass
2	50.27	2 ,4 ,33.51	Same	Pass
3	12.57	1 ,2, 4.19	same	Pass

Algorithm

1. Start.
2. Read the surface area.
3. Calculate the radius using the formula.
4. Calculate the diameter.
5.Calculate the volume.
6. Display the radius, diameter, and volume.
7. Stop.

Flowchart

Start

Input Surface Area

Calculate Radius

Calculate Diameter

Calculate Volume

Display Results

Stop

C Program

```
#include <stdio.h>
#include <math.h>
#define PI 3.14159265359
int main() {
float area, r, d, v;
printf("Enter surface area: ");
scanf("%f", &area);
```

r = sqrt(area / (4 * PI));
d = 2 * r;
v = (4.0 / 3.0) * PI * r * r * r;
printf("Radius = %.2f\n", r);
printf("Diameter = %.2f\n", d);
printf("Volume = %.2f\n", v);
return 0;
}

Compilation , Execution & Output

```
┌──(kali㉿kali)-[~]
└─$ cc sp.c -lm

┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter surface area: 314.16
Radius = 5.00
Diameter = 10.00
Volume = 523.60

┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter surface area: 50.27
Radius = 2.00
Diameter = 4.00
Volume = 33.51

┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter surface area: 12.57
Radius = 1.00
Diameter = 2.00
Volume = 4.19
```

Question 2:

Write a C program to find the image of a point after reflection in the origin.

Input, Output and Formula

Input: x-coordinate & y-coordinate
Output: Reflected point
Formula Used: Reflection about the origin: (x,y)→(−x,−y)

Standard Test Case Table

Test case	Input	Expected output	Actual output	Result
1	(2,3)	(-2,-3)	Same	Pass
2	(-4,5)	(4,-5)	Same	Pass
3	(0,-7)	(0,7)	same	Pass

Algorithm

1. Start.
2. Read x and y.
3. Multiply both coordinates by -1.
4. Display the reflected point.
5. Stop.

Flowchart

Start

Input x and y

Calculate (-x,-y)

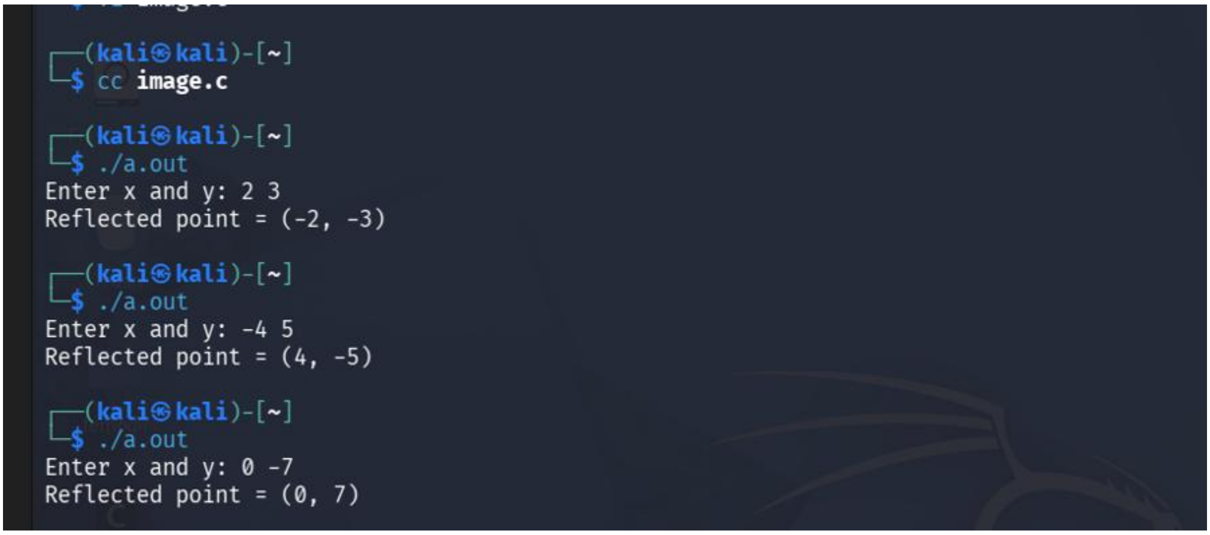
Display Reflected Point

Stop

C Program

```
#include <stdio.h>
int main()  {
int x, y;
printf("Enter x and y: ");
scanf("%d %d", &x, &y);
printf("Reflected point = (%d, %d)\n", -x, -y);
return 0;
}
```

Output


┌──(kali㉿kali)-[~]
└─$ cc image.c
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter x and y: 2 3
Reflected point = (-2, -3)
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter x and y: -4 5
Reflected point = (4, -5)
┌──(kali㉿kali)-[~]
└─$ ./a.out
Enter x and y: 0 -7
Reflected point = (0, 7)